

telemetry_processing

1.0.0-MVP — Telemetry Processing & Health

Field	Value
Revision	2
Created	2026-06-07
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Status	active
Status summary	Specifies the server-side telemetry pipeline for 1.0.0-MVP: device event ingest → OpenTelemetry/Prometheus → fleet health, the halt-on-failure inputs the rollout-engine consumes, and the post-boot health window + auto canary-abort concept carried forward from the research (active in 1.0.1 staged rollout). Composes from the observability/Herald catalogue bricks and the NEW ota-telemetry-schema module; runs inside the modular-monolith binary (ADR-0003). Staged rollout (and thus auto canary-abort) lands in 1.0.1, not MVP — MVP ingests and surfaces health and supports one-click manual abort; auto-abort is the forward concept. Post-boot health-window duration/criteria are operator-configurable and
Issues	UNVERIFIED (no fixed figure in sources; draft phase durations are non-binding). observability/Herald public surfaces are UNVERIFIED. HelixConstitution clause numbers are UNVERIFIED.
Fixed	Rev 2: added the safety invariant that halt wins over advance when both error- and success-thresholds could trigger in the same window (safety-

Field	Value
Continuation	critical rollout-gate; guarded by the §10 mutation meta-test). Pin the ota-telemetry-schema event/metric enums + codecs before repo creation; confirm observability (OpenTelemetry/Prometheus) and Herald (alert channels) public surfaces; set health-window duration + canary success/error thresholds from MVP/1.0.1 data and record them; wire the rollout-engine telemetry-signal port (see architecture §5.1) to the health-signal output here.

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The table-of-contents requirement is mandated by HelixConstitution §11.4.61 (UNVERIFIED clause number). This document carries its ToC immediately after the metadata table.

1. Purpose and scope

This document specifies how the Helix OTA control plane **ingests device telemetry, processes it through OpenTelemetry/Prometheus, derives fleet health, and feeds the rollout-engine's halt-on-failure logic** — and carries forward the **post-boot health window + automatic canary-abort** concept from the research as the forward design (active in 1.0.1 staged rollout). [master §9]

It satisfies the operator “observability” hard guarantee: *tracking/measurement/critical-data capture to detect and report problems*. [master §1]

This logic lives in the telemetry-ingest seam, with the event/metric **schema + codecs** in the NEW ota-telemetry-schema module (shared by server + agents; **no transport, no storage**). The pipeline/export and alerting reuse observability +

Herald. It runs inside the single modular-monolith binary at MVP and is the **second split candidate** (after rollout-engine) if a scale trigger fires. [architecture §4, §9; submodule-reuse-map §3/§4]

MVP boundary: MVP ingests events, drives OpenTelemetry/Prometheus, surfaces dashboard health, and supports manual (one-click) abort. The **automatic** canary-abort tied to a post-boot health window is the forward concept that activates with the staged rollout engine in **1.0.1**, because MVP deploys all-at-once (the staged engine lands 1.0.1). [master §1 non-goals, §5, §8]

2. Event ingest

Devices report a status event stream over the locked transport (HTTPS poll/report; HTTP/3→HTTP/2, REST /api/v1; control-plane JSON is Brotli/gzip-compressed — telemetry is control-plane traffic, not artifact traffic). [adr-0004 §1, §4; master §9]

The device event lifecycle (from the master design + drafts): [master §9; drafts initial_research §5.3]

download_started → installing → installed → verifying → (success | failure)

Each event carries at least: {device_id, deployment_id, status, progress, error_code/message, system_health, timestamp} (schema owned by ota-telemetry-schema; the draft StatusReport shape is a design reference). [drafts initial_research_02 §5; master §9]

Ingest responsibilities (the telemetry-ingest seam): - Authenticate the reporting device (device token bound to hardware id; auth/security). [master §6] - Validate the event against the ota-telemetry-schema codec; reject malformed events (no silent acceptance). - Normalize and forward to the processing pipeline (§3); persist raw events / derived metrics in database (telemetry_events / update_metrics). [master §7] - Emit a **health signal** consumable by the rollout-engine's telemetry-signal port (§5). [architecture §5.1]

3. Processing pipeline (ingest → OTel/Prometheus → health)

graph LR

```
AGENT["Android agent (KMP)"] -->|"HTTPS poll/report (REST /  
api/v1, Brotli)"| ING["telemetry-ingest (seam)"]  
ING --> SCHEMA["ota-telemetry-schema (validate/decode)"]  
ING --> DB[(database: telemetry_events / update_metrics)]  
ING --> OTEL["observability (OpenTelemetry → Prometheus)"]  
OTEL --> HEALTH["fleet health model (§4)"]  
HEALTH --> ROLL["rollout-engine telemetry-signal port (§5)"]  
HEALTH --> HERALD["Herald (alerts §7)"]  
HEALTH --> DASH["Dashboard health view"]
```

The pipeline maps directly to master §9: *device event stream* → *ingest* → *OpenTelemetry/Prometheus* → *dashboard health + alerting via Herald*; *metrics drive the rollout halt logic (§8) and problem detection/reporting*. [master §9]

4. Fleet health model

Health is computed **per deployment** (and per phase/cohort in 1.0.1) from the event stream: [master §8, §9]

Metric	Derived from	Used by
Success rate	count(success) / count(devices in cohort that reached a terminal state)	rollout success-threshold check (§5)
Error/failure rate	count(failure) / count(cohort terminal)	rollout error-threshold check (§5); alerting (§7)
In-flight progress	distribution over download_started/ installing/ installed/verifying	dashboard; stuck-device detection
Post-boot health	verifying → success confirmation within the health window (§6)	canary-abort decision (§6, 1.0.1)
System health	device-reported system_health field	problem detection/ reporting (operator requirement)

Health is the **single source of truth** for both the rollout halt decision (§5) and alerting (§7). It is derived from Prometheus metrics produced via observability/OpenTelemetry, not recomputed ad hoc. [master §9]

5. Halt-on-failure inputs (rollout-engine contract)

The rollout-engine is OS-agnostic and **HTTP-free**; it consumes telemetry **signals** through a port (it contains no ingest itself). This section defines the signals telemetry processing supplies. [architecture §5.1; master §8; submodule-reuse-map §4]

The engine config is ordered phases {percentage, success_threshold, error_threshold, duration, auto_progress}; it **starts a phase, targets the phase cohort, monitors telemetry, halts/pauses on error-threshold breach, advances on success-threshold within duration**. [master §8]

Signals supplied to the engine's telemetry-signal port:

Signal	Meaning	Engine decision
error_rate ≥ error_threshold (phase cohort)	failures exceed the phase's tolerance	halt/pause (and alert §7)

Signal	Meaning	Engine decision
success_rate ≥ success_threshold within duration	phase met its bar in time	advance to next phase
success_rate < success_threshold at duration expiry	phase did not meet its bar	hold/pause (operator decision)
post_boot_health_failed (cohort, within window §6)	canary devices unhealthy after boot	abort (auto in 1.0.1; manual at MVP)

Safety invariant — halt wins over advance. If, within the same evaluation window, both the error-threshold (`error_rate ≥ error_threshold`) and the success-threshold (`success_rate ≥ success_threshold`) conditions could trigger, the engine **MUST** take the **halt/pause** decision, never **advance**. The rollout-gate is safety-critical: when in doubt, stop. This precedence is a hard rule the gate enforces and the mutation meta-test (§10 layer 4) guards — an implementation that advances despite a concurrent error-threshold breach is a failure.

The signal contract is identical whether the rollout-engine fronts hawkBit (wrap) or is the Go-native engine (fallback) — the wrap/fallback choice (ADR-0001) does not change what telemetry must supply. [architecture §5.1; adr-0001 §4] The rollout-gate is **safety-critical (≥90% coverage)**, so these signals and their thresholds are deterministically testable (fake clock + fake telemetry). [master §8, §13; adr-0003 §6]

MVP note: MVP deploys all-at-once, so the per-phase signals above are produced and surfaced, but **automatic** halt/advance is a 1.0.1 staged-rollout capability; at MVP these drive **dashboard health** + **manual abort**. [master §5, §8]

6. Post-boot health window + auto canary-abort

This is the forward concept carried from the research drafts and the master design, made explicit here. [drafts initial_research §5.3; drafts initial_research_02 §6; master §8, §9]

Post-boot health window. After a device applies an update and reboots, it reports verifying and runs a **post-boot health check**; on success it reports success, otherwise failure. [drafts initial_research §5.3] The **window** is the bounded period during which a freshly-updated device must confirm health (verifying → success) before it is counted healthy. If the device does not confirm within the window, it is treated as a **post-boot failure** for the cohort’s health math (§4). The **window duration and success criteria are operator-configurable; no fixed figure is asserted** — the drafts’ phase durations are explicitly non-binding. [adr-0003 §3.2 (draft constants non-binding); master §8] On-device, this layers on top of update_engine’s own guarantee: update_verifier marks the slot successful, and a failed boot triggers **automatic A/B rollback** independently of the server. [drafts initial_research_02 §1; master §1, §6]

Auto canary-abort (1.0.1). The first rollout phase is the **canary** (smallest cohort, e.g. the draft’s 5% “Canary” with success_threshold/error_threshold). [drafts initial_research_02 §6] If, within the post-boot health window, the canary cohort’s

post-boot failure/error rate breaches the phase's `error_threshold`, the rollout-engine **auto-aborts the rollout** (`pause_on_error` / `rollback_on_critical_failure` in the draft strategy) and raises an alert (§7), rather than advancing to wider cohorts. [drafts `initial_research_02` §6; master §8] This is the Memfault-style “server-side one-click abort” made automatic by the health window — the pattern ADR-0001 names for the Go-native engine. [adr-0001 §3.3]

sequenceDiagram

```
participant Cohort as Canary cohort
participant TEL as telemetry-ingest
participant H as health model
participant RE as rollout-engine
Cohort->>TEL: verifying / success / failure (post-boot)
TEL->>H: events within health window
H->>RE: post_boot_health_failed? (rate vs error_threshold)
alt within window, breach
    RE->>RE: ABORT (auto in 1.0.1; manual at MVP)
    RE->>TEL: emit alert (Herald §7)
else healthy within window
    RE->>RE: advance to next phase
end
```

MVP vs 1.0.1: at **MVP** the post-boot health window is **observed and surfaced** (events ingested, health computed, dashboard + manual one-click abort available); **automatic** canary-abort activates with the **1.0.1** staged rollout engine. [master §1 non-goals, §5, §8]

UNVERIFIED: window duration, canary cohort size, and success/error thresholds — no binding figures exist in the sources (draft constants are non-binding); set from data and record. [adr-0003 §3.2; master §8]

7. Alerting & notifications

Health breaches and aborts dispatch alerts via **Herald** (alert routing) with internal domain-event fan-out via **eventbus**; the control plane **publishes** events and does not implement a notifier. [master §9; submodule-reuse-map §3] Triggers: error-threshold breach (§5), post-boot health-window failure / canary-abort (§6), and stuck/unreporting devices (§4). Alert channel coverage by Herald is **UNVERIFIED**. [submodule-reuse-map §3]

8. MVP vs 1.0.1 boundary

Capability	1.0.0-MVP	1.0.1 (staged rollout)
Event ingest (§2)	yes	yes
OpenTelemetry/ Prometheus + dashboard health (§3, §4)	yes	yes
	yes	yes

Capability	1.0.0-MVP	1.0.1 (staged rollout)
Persist telemetry_events / update_metrics (§2)		
Halt-on-failure signals computed (§5)	yes (surfaced)	yes (drive engine)
Automatic halt/advance per phase (§5)	no (all-at-once)	yes
Post-boot health window observed (§6)	yes	yes
Automatic canary-abort (§6)	no (manual one-click)	yes
Alerting via Herald (§7)	yes	yes

This boundary follows the master non-goals (staged rollout + automatic phase control land 1.0.1) while keeping MVP fully observable and manually controllable. [master §1, §5, §8]

9. Catalogue-first composition

Concern	Brick / module	Class
Event/metric schema + codecs	ota-telemetry-schema	new
Telemetry pipeline / export (OpenTelemetry → Prometheus)	observability	reuse (UNVERIFIED surface)
Alert routing	Herald	reuse (UNVERIFIED surface)
Internal domain-event fan-out	eventbus	reuse
Event + derived-metric persistence	database (PostgreSQL)	reuse
Device auth on report	auth, security	reuse

The schema module carries **no transport, no storage**; only canonical catalogue names used (none invented). [submodule-reuse-map §3/§4]

10. Testing (four-layer)

The rollout-gate fed by these signals is **safety-critical (≥90% coverage)** with mutation immunity. [master §13; adr-0003 §6]

Layer	What it asserts for telemetry processing
1. Source-presence gate	ota-telemetry-schema defines the event/metric enums + codecs; the

Layer	What it asserts for telemetry processing
2. Artifact gate	telemetry-ingest seam exposes the health-signal port; observability/Herald/eventbus/database are wired (not re-implemented); health-window + threshold config keys exist. The telemetry path ships in the control-plane binary; ingested events produce Prometheus metrics via observability (assert metrics are emitted); persisted telemetry_events/update_metrics rows are written; telemetry travels as Brotli/gzip control-plane JSON (never on the artifact identity path). Feed a synthetic event stream (download_started...success/failure); assert success/error rates compute correctly; cross an error-threshold and assert a halt/pause signal is produced and an alert is dispatched; simulate post-boot health-window expiry without success and assert post_boot_health_failed; (1.0.1) assert auto canary-abort fires on canary breach within the window. Use a fake clock + fake telemetry for determinism.
3. Runtime / integration	Negate a halt invariant (e.g. let the engine advance despite an error-threshold breach, or count a window-expired device as healthy) and assert the rollout-gate test flips PASS→FAIL. The safety of halt-on-failure / canary-abort is proven by the negation breaking the gate.
4. Mutation meta-test	

No-bluff positive evidence only (§7.1, UNVERIFIED). [adr-0003 §6] On-device, the post-boot/A-B-rollback path is validated on the real Orange Pi 5 Max plan (download→verify→apply→reboot→verify; corrupt-slot→confirm A/B fallback). [master §13]

11. Compliance notes (HelixConstitution)

Clause numbers carried from the corpus convention; **UNVERIFIED** against the authoritative text. [submodule-reuse-map §7]

Clause	How telemetry processing complies
§1 observability guarantee	Event ingest → OTel/Prometheus → health detects and reports problems;

Clause	How telemetry processing complies
	alerting via Herald (§3, §4, §7). [master §1, §9]
§11.4.28 (decoupling)	Schema (ota-telemetry-schema) is transport/storage-free; ingest supplies signals to the rollout-engine via a port; engine contains no ingest (§2, §5). [architecture §5.1]
§11.4.74 (catalogue-first)	Pipeline/alerting reuse observability/Herald/eventbus; only the schema is new (§9). [master §10]
§11.4.6 / §11.4.8 (no-guessing / research-first)	Health-window/threshold figures carried UNVERIFIED (draft constants non-binding); MVP-vs-1.0.1 boundary set on the master non-goals, not invented (§6, §8). [adr-0003 §3.2]
§1 / §1.1 (four-layer + mutation)	§10 testing; the halt-on-failure/canary-abort path floored $\geq 90\%$ with mutation immunity. [master §13]
§11.4.123 (rock-solid proof)	Halt/abort behaviour proven by negation flipping the gate (§10 layer 4) + on-device A/B-fallback validation, not by assertion. UNVERIFIED against clause text. [master §13]

12. Open / UNVERIFIED items

1. **Health-window duration, canary cohort size, success/error thresholds** — no binding figures in sources (draft constants non-binding). **UNVERIFIED**. [adr-0003 §3.2; master §8]
2. **observability (OpenTelemetry/Prometheus) and Herald (alert channels) public surfaces** — not inspected this revision. **UNVERIFIED**. [submodule-reuse-map §7]
3. **ota-telemetry-schema event/metric enums + codecs** — to pin before repo creation. [master §10]
4. **Telemetry-vs-rollout volume asymmetry** (telemetry plausibly higher-volume → second split candidate) — **UNVERIFIED**; confirm via MVP load tests. [adr-0003 §3.2; architecture §9]
5. **Constitution clause numbers** — carried from corpus convention. **UNVERIFIED**.

13. Sources

All paths relative to docs/research/main_specs/.

- 00-master/2026-06-07-helix-ota-design.md — §1 observability guarantee + non-goals, §5 MVP flow, §8 staged rollout engine, §9 telemetry & observability, §13 testing.

- research/adr/adr-0003-server-topology.md — telemetry as second split candidate, draft constants non-binding, rollout-gate $\geq 90\%$ floor.
- research/adr/adr-0004-transport.md — telemetry is control-plane traffic (Brotli/gzip), distinct from the artifact identity path.
- research/adr/adr-0001-wrapped-engine.md — Memfault one-click-abort pattern; signal contract identical across wrap/fallback.
- 00-master/submodule_reuse_map.md — ota-telemetry-schema boundary; observability/Herald/eventbus bindings.
- additions/initial_research.md — post-boot health check + recovery flow (device lifecycle).
- additions/initial_research_02.md — canary phase + rollout strategy config (pause_on_error, rollback_on_critical_failure); update_verifier slot-success.
- architecture.md — the telemetry-ingest seam and rollout-engine telemetry-signal port this spec feeds.